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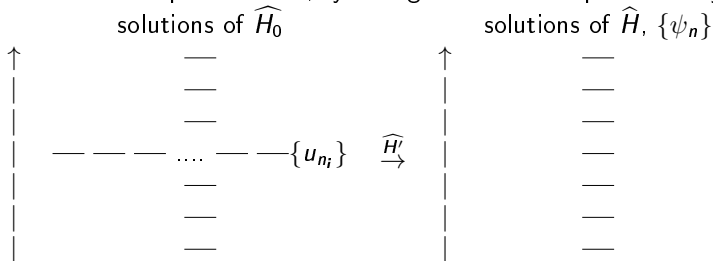
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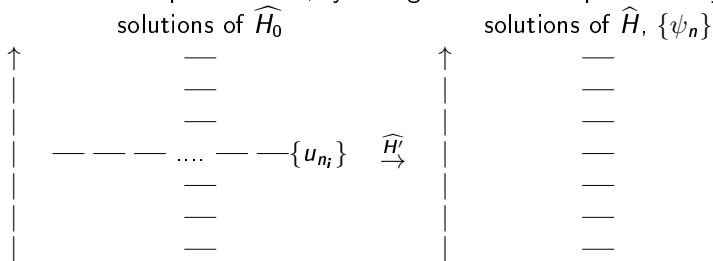
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-degenerate states go back to a linear combination of the original wavefunctions;
 so it is more complicated than the non-degenerate case
 consider here case of Non-degenerate levels only

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then the second term = $\langle u_m | -\Delta E_n | \psi_n \rangle = -H'_{nn} \langle u_m | u_n \rangle \stackrel{m \neq n}{=} 0$.

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1st order correction to ground state energy:

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$$x = \sqrt{\frac{\hbar}{2\mu\omega}} (a + a^\dagger); \quad \therefore V_{11} = \sqrt{\frac{\hbar}{2\mu\omega}} \langle 0 | a + a^\dagger | 0 \rangle = 0$$